

5/5/2020

SIGNAL & DATA ANALYSIS IN NEUROSCIENCE 2020

NEURAL DECODING & PARAMETERS ESTIMATION

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Outline

- Decoding
 - Population vector
- Model based (parametric) estimation
 - Bayesian estimator
 - MAP estimator
 - ML Estimator

Population vector - exam 2010

- The Green Grass Goblin has four neurons for identifying Elf movements with different optimal directions. The neurons fired at the following rates for an Elf approaching from direction X:

Optimal direction	0	90	180	270
Firing rate for X	20	10	4	8

All the neurons have a max FR of 30 spikes/s and baseline FR is 0 sp/sec.

Calculate the movement direction of the Elf using a population vector.

$$\vec{v}_{population} = \sum_{i=1}^N \frac{r_i - r_0}{r_{max}} \cdot \vec{v}_i$$

5/5/2020

4

Solution

- Unit vectors for optimal directions: ($r=1$)
 $x = 1 \cdot \cos(\alpha)$, $y = 1 \cdot \sin(\alpha)$

	x	y
0:	[1 0]	
90:	[0 1]	
180:	[-1 0]	
270:	[0 -1]	
- Population vector:
[20/30 10/30 4/30 8/30] * [1 0]
[0 1]
[-1 0]
[0 -1] = [0.5333 0.067] (X, Y)
 $\Rightarrow \text{Theta} = \tan^{-1}(y/x) = 7.1 \text{ deg.}$

5

Parameters estimation

- Suppose we know/assume a given model for the data, and we get a sample of observations
 $X = \{x_1 \dots x_n\}$
- We don't know the parameters of the model
- Use Bayes to estimate the parameters

6

Estimation methods

- Maximum a posteriori estimator (MAP):**
$$\hat{\Theta} = \operatorname{argmax}_{\Theta} P(\Theta|x)$$
$$p(\Theta|x) = \frac{p(x|\Theta) \cdot p(\Theta)}{p(x)}$$
$$\hat{\Theta} = \operatorname{argmax}_{\Theta} P(x|\Theta) \cdot p(\Theta)$$
- Maximum Likelihood Estimator (MLE):**
 - Estimating model parameters given an observation – similar to MAP without the prior...
$$\hat{\Theta} = \operatorname{argmax}_{\Theta} P(x|\Theta)$$
- Bayesian approach:**
Given cost function $C(\epsilon = \Theta - \hat{\Theta})$ and x :
$$\hat{\Theta} = \operatorname{argmin}_{\hat{\Theta}} \int_{\Theta} C(\Theta, \hat{\Theta}) \cdot p(\Theta|x) d\Theta$$

7

Example MAP: Exam 2007

- The time difference (in weeks) between occurrences of amnesia in Foergetis Homeworkis patients may generally be estimated by an exponential distribution:

$$p(x) = \begin{cases} \lambda \cdot e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The parameter λ describing the distribution is different for each patient. The five observations of amnesia in patient X are spaced by 2, 1, 5, 1, 3 weeks.

Assuming the prior distribution $P_0(\lambda) = \frac{1}{2} \lambda$, ($0 \leq \lambda \leq 2$) known for the general patient population, what is the maximum a-posteriori (MAP) estimator?

8

Solution

Given: $p(x) = \lambda \exp(-\lambda x)$, $\underline{X} = [1, 1, 2, 3, 5]$, $p(\lambda) = 0.5(\lambda)$
Find: MAP for λ

$$\begin{aligned} \hat{\lambda} &= \operatorname{argmax}_{\lambda} P(\lambda | \underline{X}) \quad / \text{ MAP} \\ P(\lambda | \underline{X}) &= \frac{P(\underline{X} | \lambda) \cdot P(\lambda)}{P(\underline{X})} \quad / \text{ Bayes} \\ F = \prod_i P(x_i | \lambda) \cdot P(\lambda) &= \lambda^5 \exp^{-12\lambda} \cdot \frac{\lambda}{2} = \frac{1}{2} \lambda^6 \exp^{-12\lambda} \quad / x_i \text{ iid} \\ \frac{dF}{d\lambda} &= \frac{1}{2} \lambda^5 \exp^{-12\lambda} [6 - 12\lambda] \\ \lambda_1 = 0 & \quad P(\lambda_1 | \underline{X}) = 0 \\ \lambda_2 = \frac{1}{2} & \quad P(\lambda_2 | \underline{X}) > 0 \text{ and max point,} \\ \text{Hence, } \hat{\lambda} &= \frac{1}{2} \end{aligned}$$

9

Example MLE: Exam 2005

- During 5 trials the recorded rates of the neuron were: 70, 62, 95, 59, 65. Assuming that the recorded rate is the result of a Poisson distribution, find the MLE for the parameter λ .

5/5/2020

10

Solution

Given: $X \sim \text{Poisson distribution}$. $X = [59, 62, 65, 70, 95]$
 Find: MLE λ

$$\hat{\lambda} = \operatorname{argmax}_{\lambda} P(\underline{X}; \lambda) \text{ / MLE}$$

$$f(x_i; \lambda) = \frac{\lambda^{x_i} \exp^{-\lambda}}{x_i!} \text{ / Poisson distribution}$$

$$f(\underline{X}; \lambda) = \prod_i \frac{\lambda^{x_i} \exp^{-\lambda}}{x_i!}$$

$$\ln(f(\underline{X}; \lambda)) = \sum_i \ln \left(\frac{\lambda^{x_i} \exp^{-\lambda}}{x_i!} \right)$$

$$F = \sum_i \ln(\lambda^{x_i} \exp^{-\lambda}) = \sum_i x_i \ln(\lambda) - \lambda \cdot N$$

$$\frac{dF}{d\lambda} = \frac{\sum_i x_i}{\lambda} - N = 0$$

$$\lambda = \frac{\sum_i x_i}{N} \text{ / max point}$$

$$\hat{\lambda} = E[x_i]$$

11

Example: MAP +MLE exam 2006

- The time difference (in hours) between occurrences of hiccups in Hiccupitis Neuralitis Syndrome patients may be estimated by the following uniform distribution:
 $p(x) = 1/n$ $0 \leq x \leq n$
 $p(x) = 0$ $x > n$
- The parameter n describing the distribution is different for each patient. The five observations of hiccups in patient X are spaced by 2, 3, 7, 2, 1 hours.
- Find the maximum likelihood (ML) estimator for n in patient X and explain the results.
- Assuming the priors $P_0(7)=0.1$ & $P_0(8)=0.9$ known for the general patient population. Is the maximum aposteriori (MAP) estimator different from the ML estimator calculated in section (a)?

12

Solution

Given: $P(x_i) = 1/n$, $X = [1, 2, 2, 3, 7]$
 Find: MLE for $\hat{n} = \operatorname{argmax}_n P(\underline{X}|n)$
 Solution: $= \prod_i P(x_i|n) = \left(\frac{1}{n}\right)^N$

(i) $n \geq 7$
 (ii) Minimal n will maximize $P(\underline{X}|n)$
 $n = 7$

5/5/2020

13

Solution cont

Given prior for n, find MAP for n

Solution:

$$\hat{n} = \operatorname{argmax}_n P(n|\underline{X})$$

$$P(n|\underline{X}) = \frac{P(\underline{X}|n) \cdot P(n)}{P(\underline{X})}$$

$$F = P(\underline{X}|n) \cdot P(n) = \prod_i P(x_i|n) \cdot P(n)$$

$$F = \left(\frac{1}{n_0}\right)^N \cdot P(n = n_0)$$

$$F(n=7) = \left(\frac{1}{7}\right)^5 \cdot 0.1 = 6e^{-6}$$

$$F(n=8) = \left(\frac{1}{8}\right)^5 \cdot 0.9 = 2.7e^{-5}$$

$$\hat{n} = 8$$
